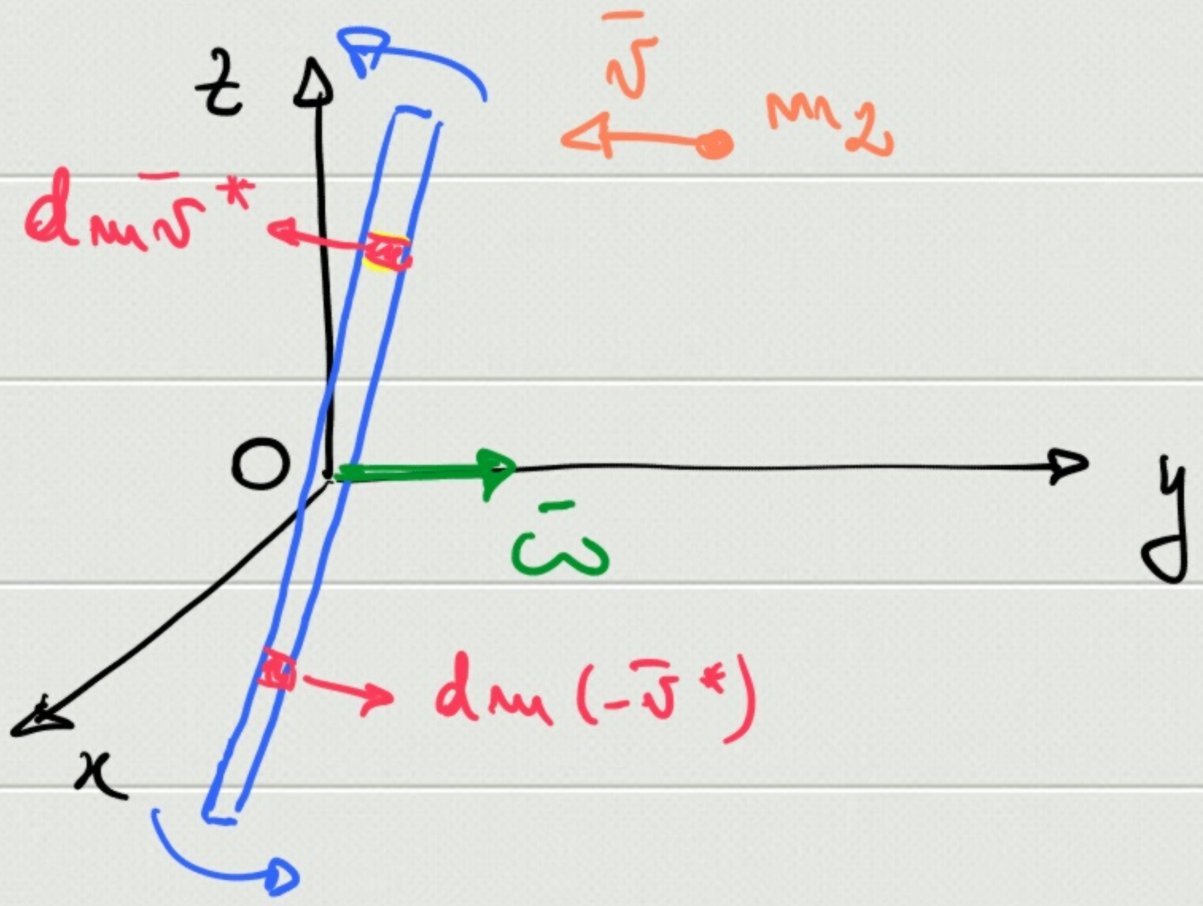


m_1, ℓ

$\bar{v} = v(-\bar{u}_y)$

urto compl. anelastico

$$\bar{P}^- = \bar{P}^-_{sbarretta} + \bar{P}^-_{m_2} = m_2\bar{v} = m_2 v(-\bar{u}_y)$$

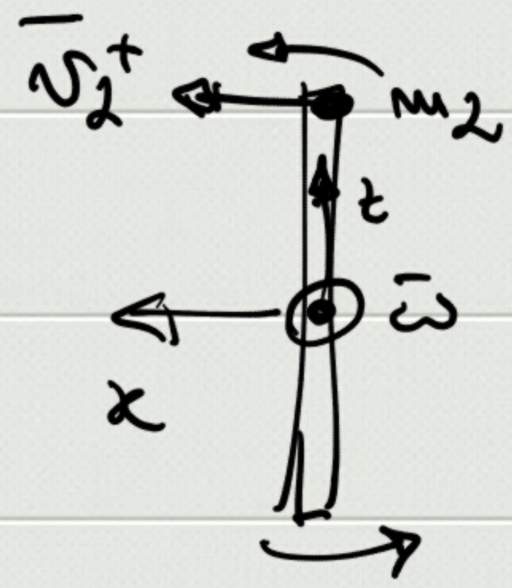
$\bar{P}^-_{m_2} \to m_2\bar{v}$

$\hookrightarrow m_{sb}\,\bar{v}_{CM,sb} = 0$

$$\bar{P}^+ = \bar{P}^+_{sb} + \bar{P}^+_{m_2} = m_2\bar{v}_2^+ = m_2 v_2^+(\bar{u}_x) =$$

$\hookrightarrow = 0$

$$= m_2\omega'\frac{\ell}{2}(\bar{u}_x)$$



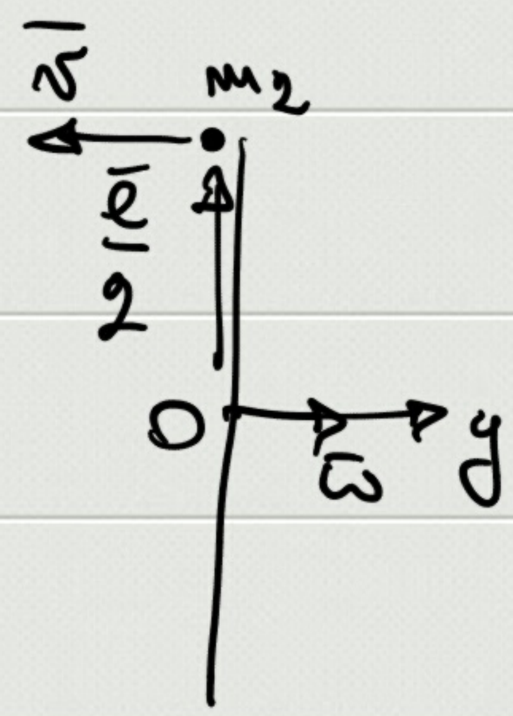
$$\Delta\bar{P} = \bar{P}^+ - \bar{P}^- \neq 0 \qquad \boxed{\bar{P} \neq cost}$$

$$\bar{J} = \Delta \bar{P} = m_2 \omega' \frac{\ell}{2} \bar{u}_x - m_2 \bar{v} (-\bar{u}_y) =$$

$$= m_2 \left(\omega' \frac{\ell}{2} \bar{u}_x + v \bar{u}_y \right)$$

$$\Rightarrow \quad J = m_2 \sqrt{\left(\omega' \frac{\ell}{2}\right)^2 + v^2}$$

$$\bar{L}_o^- \quad , \quad \bar{L}_o^+$$



$$\bar{L}_o^- = \bar{L}_{o,sb}^- + \bar{L}_{o,m_2}^- =$$

$$= \bar{I}_y \bar{\omega} + \frac{\bar{\ell}}{2} \times m_2 \bar{v} =$$

$$= \underbrace{\frac{1}{12} m_1 \ell^2 \omega \bar{u}_y}_{} + \frac{\ell}{2} m_2 v \bar{u}_x$$

$$\bar{L}_o^+ = \bar{L}_{o,sb}^+ + \bar{L}_{o,m_2}^+ = \quad \left(= \bar{L}_{o,sb+m_2}^+ \right) \searrow \bar{I}_y' \bar{\omega}'$$

$$= \bar{I}_y \bar{\omega}' + \frac{\bar{\ell}}{2} \times m_2 \bar{v}_2^+ =$$

$$= \frac{1}{12} m_1 \ell^2 \bar{\omega}' + \frac{\ell}{2} m_2 \omega' \frac{\ell}{2} \bar{u}_y =$$

$$= \underbrace{\frac{\ell^2}{4} \left(\frac{1}{3} m_1 + m_2 \right) \omega' \bar{u}_y}_{} \quad \Rightarrow \boxed{\bar{L}_o \neq \text{cost}}$$

$L_{o,x} \neq \text{cost}$ $\qquad$ $(L_{o,z} \neq \text{cost})$

$$\boxed{L_{o,y} = \text{cost}}$$

$$\Rightarrow \frac{1}{\cancel{12}_{3}} m_1 \cancel{\ell^2} \omega \bar{v}_y = \frac{\cancel{\ell^2}}{4} \left(\frac{1}{3} m_1 + m_2 \right) \omega' \bar{v}_y$$

$$\Rightarrow \boxed{\omega' = \frac{m_1}{m_1 + 3m_2} \omega}$$

$$\int \bar{M} \, dt = \Delta \bar{L}$$

$$\Rightarrow \Delta \bar{L}_o = \bar{L}_o^+ - \bar{L}_o^- =$$

$$= \cancel{\left(L_{oy}^+ \bar{v}_y \right)} - \left(L_{ox}^- \bar{v}_x + \cancel{L_{oy}^- \bar{v}_y} \right) =$$

$$= - L_{ox}^- \bar{v}_x = - \frac{\ell}{2} m_2 v \bar{v}_x$$